Leveraging higher-order time integration methods for improved computational efficiency in a rainshaft model

Scientific Achievement

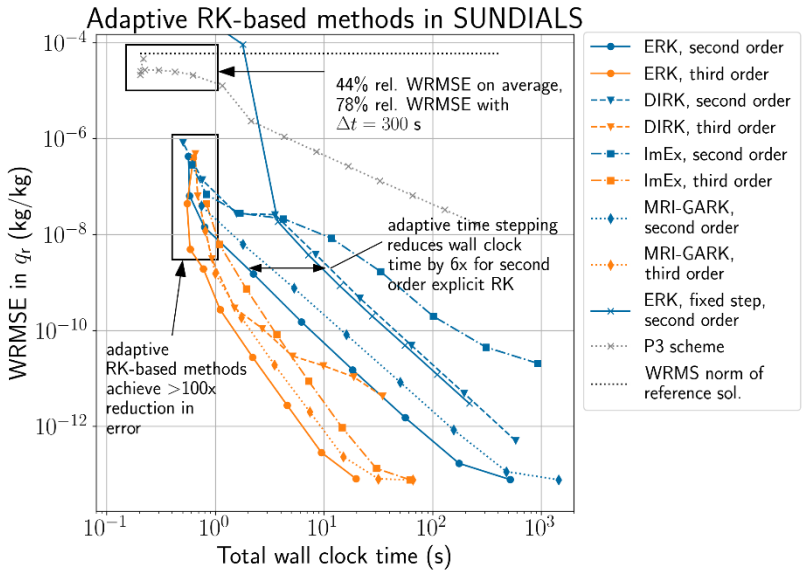
We improved the accuracy and efficiency of cloud microphysics schemes like those currently used in the Energy Exascale Earth System Model (E3SM) by utilizing high-order time integration methods with adaptive time stepping.

Significance and Impact

- Accurate simulation of cloud microphysics is essential for understanding the production of precipitation in clouds and the water cycle.
- Our work demonstrates that current time integrators for cloud microphysics use large time steps that are inadequate for resolving key precipitation processes.
- This collaboration between atmospheric scientists and mathematicians shows that adaptive, high-order time integration methods resolve precipitation processes 10x faster than current schemes without the need for *ad hoc* limiters.

Technical Approach

- Microphysical processes were analyzed to determine their associated timescales and stability properties, which helped inform us which time integration methods were likely to perform best.
- Five classes of time integrators were implemented using the FASTMath SUNDIALS (Suite of Nonlinear and Differential/ALgebraic equation Solvers) in SPAECIES, a new software framework we developed that allows users to integrate microphysical processes in arbitrary groupings and orderings using SUNDIALS.



Total wall clock time in seconds vs. weighted root mean square error estimate (WRMSE) in the rain mass mixing ratio using the default time integration scheme in E3SM (P3 scheme) vs. suite of Runge–Kutta-based time integration methods in SPAECIES. Second and third-order explicit Runge–Kutta methods with adaptive time step control perform best, reducing the WRMSE by a factor of 100 and resolving key precipitation features 10x faster.

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